PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Repetitive transcranial magnetic stimulation for upper limb motor function and activities of daily living in stroke patients: A protocol of a systematic review and Bayesian network meta-analysis
AUTHORS	Lu, Yue; Xia, Yuan; Wu, Yue; Pan, Xinyong; Wang, Zhenyu; Li, Yongjie

VERSION 1 – REVIEW

REVIEWER	Zewdie, Ephrem University of Calgary, Department of Pediatrics
REVIEW RETURNED	24-Jul-2021

GENERAL COMMENTS	1. The manuscript requires heavy grammar and spelling editing. 2. In the introduction, the description of single-pulse and paired-pulse TMS is wrong. These two terms are not used to describe repetitive TMS (rTMS). Please review. 3. In your methods section, you described your control group to be a 'routine treatment group'. It is not clear what a routine treatment is. Please describe clearly. 4. Spasms and spasticity are used interchangeably while they don't mean the same thing. Please make sure to use the proper terms. 5. Although you are including the type of the stroke as a factor, make sure to include the location and extent of stroke as variables that will affect your analysis. 6. Overall, I look forward to reading the outcome of the review. As a protocol paper, it is not adding any knowledge or using any unique methods. Before the full review is completed, I do not think this is sufficient for publication.

REVIEWER	Xie, Qing Shanghai Jiao Tong University
REVIEW RETURNED	10-Aug-2021

GENERAL COMMENTS	This paper is a meta-analysis method to demonstrate the efficacy of different rTMS interventions in upper limb motor function and activities of daily living in stroke patients. This protocol has reasonable idea and design, and can meet the practical clinical need. However, the grammar and English word usage are unclear which limits the usefulness of the information in the manuscript. Please have the manuscript reviewed and edited by someone who is proficient in the use of professional English. Also, there are more details needed to be clearly described:

	1. If the design innovation of this protocol can be better expressed, it will make the article more valuable. 2. intervention site To add the intervention site factor or directly exclude the intervention site except M1 are my suggestion. The intervention site cannot be ignored although the author had mentioned that the modality of the intervention is the more important compared to another factor.. 3. the modality of intervention Based on the Interhemispheric inhibition model, the general intervention is low frequency of M1 on the unaffected hemisphere and high frequency of M1 on the affected hemisphere. However, some studies have shown that high frequency on the healthy hemisphere is also effective in patients with severe dysfunction. It would be more comprehensive if author can consider the modalities which beyond the Interhemispheric inhibition model.

VERSION 1 – AUTHOR RESPONSE

Comments from Reviewer
-Reviewer 1
Comment 1
- The manuscript requires heavy grammar and spelling editing.
Response
We have got our manuscript edited/ proofread from an expert scientific writer in the field of biomedical sciences. The language of revised manuscript is now highly improved with good scientific organization.

Comment 2
- In the introduction, the description of single-pulse and paired-pulse TMS is wrong. These two terms are not used to describe repetitive TMS (rTMS). Please review.
Response
We appreciate the reviewer for this suggestion. We revisited the literature and found that there were errors in the description of single-pulse and paired-pulse TMS in the introduction. Therefore, we re-described the relevant contents of rTMS in the second paragraph of the introduction. Changes made in the revised manuscript file are highlighted in RED.
Comment 3
-In your methods section, you described your control group to be a 'routine treatment group'. It is not clear what a routine treatment is. Please describe clearly.
Response
We have further emphasized the content of the routine treatment group in "Types of interventions" of the method.

Comment 4
-Spasms and spasticity are used interchangeably while they don't mean the same thing. Please make sure to use the proper terms.
Response
We appreciate the reviewer for this suggestion. "Spasms" has been replaced with "spasticity" in "Types of interventions" of the method. Changes made in the revised manuscript file are highlighted in RED.
Comment 5

- Although you are including the type of the stroke as a factor, make sure to include the location and extent of stroke as variables that will affect your analysis.
Response
Since location and extent of stroke as variables that will affect our analysis, we have added the two as factors in the subgroup analysis to the second paragraph of statistical analysis. Changes made in the revised manuscript file are highlighted in RED.
Comment 6
- Overall, I look forward to reading the outcome of the review. As a protocol paper, it is not adding any knowledge or using any unique methods. Before the full review is completed, I do not think this is sufficient for publication.
Response
As a protocol paper, our study does have certain limitations. We have comprehensively revised the manuscript based on your comments to meet the publication requirements as much as possible. Taking into account the scientific of the research, the methods involved in our study refer to the previous protocol paper published by BMJ OPEN. In addition, the results of our review are being completed. When we submit the article, I will invite you as a reviewer and look forward to your valuable comments.

Reviewer 2
Comment 1
- Please have the manuscript reviewed and edited by someone who is proficient in the use of professional English. If the design innovation of this protocol can be better expressed, it will make the article more valuable.
Response
We appreciate the reviewer for this suggestion. We have got our manuscript edited/ proofread from an expert scientific writer in the field of biomedical sciences. The language of revised manuscript is now highly improved with good scientific organization. In addition, we have optimized the expression of design innovation in the third paragraph of the introduction. Changes made in the revised manuscript file are highlighted in RED.

Comment 2
intervention site
- To add the intervention site factor or directly exclude the intervention site except M1 are my suggestion. The intervention site cannot be ignored although the author had mentioned that the modality of the intervention is the more important compared to another factor.
Response
We have directly excluded the intervention site except M1 are. Meanwhile, we have added the above contents in our revised manuscript in “Exclusion criteria” of the method. Changes made in the revised manuscript file are highlighted in RED.

Comment 3
the modality of intervention
-Based on the Interhemispheric inhibition model, the general intervention is low frequency of M1 on the unaffected hemisphere and high frequency of M1 on the affected hemisphere. However, some studies have shown that high frequency on the healthy hemisphere is also effective in patients with severe dysfunction. It would be more comprehensive if author can consider the modalities which beyond the Interhemispheric inhibition model.
Response
We appreciate the reviewer for this suggestion and have now considered "high frequency rTMS on the unaffected side" as a stimulation group. Meanwhile, we have added the above contents in our revised manuscript in “Introduction” , “span style="font-family:'Times New Roman'; font-

weight:bold">Types of interventions of the method” and “Discussion”.Changes made in the revised manuscript file are highlighted in RED.